P-521 fixed-width field validation

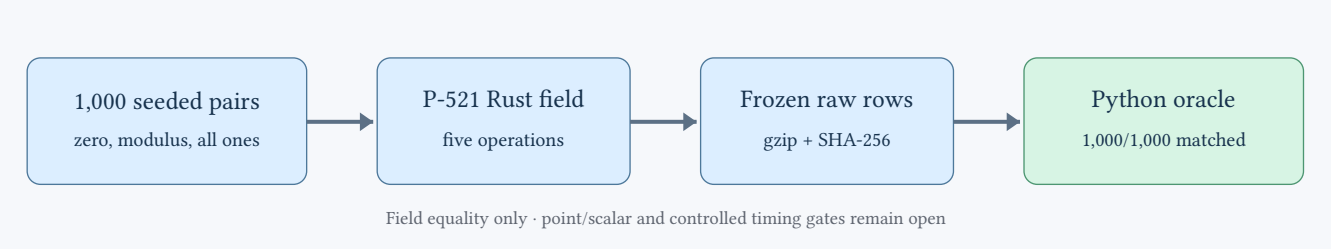
Status: local arithmetic correctness observation. No timing, point, scalar, or side-channel conclusion follows. The source hashes, exact commands, compiler stderr, and compressed raw rows are in `research/ecc_arithmetic_20261013/p521_validation/README.md`.

Implementation boundary

The new field uses nine little-endian 64-bit words, a `2^521 = 1` end-around reduction, 81 schoolbook word products per multiply, and a fixed inversion chain for `2^521 - 3`. It is exposed as a module but no P-521 point path calls it yet. The source was uncommitted relative to `9f588c4f4` during the run.

Observation

On Apple ARM64 with Rust 1.93.1, the Rust source emitted 1,000 deterministic rows for canonicalization, add, subtract, multiply, and invert. A separately written Python integer checker matched every row modulo `2^521 - 1`. The raw rows are retained compressed with a SHA-256 digest. The checker invocation before compression failed because the `.gz` file was missing; its repeat after compression passed. The same source cross-compiled to an x86-64 Linux object with Rust 1.98.0; that object was retained but did not execute on an x86-64 host. The source-bound Cargo harness passed 13 focused tests, including this module's BigUint differential test, after a retained test-only compile fix.



Remaining gates

The repository release test, compiled-code side-channel review, P-521 point integration and scalar correctness, and isolated A/A plus matched A/B timing panels remain. This diagram represents the observed field check only.